PegGuard: Stablecoin Reserve Stress Tester

Will this stablecoin still be worth $1 when everyone wants out?

Team Bulge **Track** Technical (build) **Category** Consumer or B2B fintech application

Stack Python · Streamlit · Plotly · pytest **Repo** see Canvas submission

1 What does it do?

PegGuard takes a fiat-backed stablecoin's **reserve portfolio** and simulates a **redemption run** against it, day by day, to show whether the issuer can still pay every token $1 on time.

1. **Input reserves.** The user picks a bundled portfolio (USDT and USDC compositions from their Q2 2026 reports, an SVB-style duration mismatch, the USDC/SVB weekend of March 2023, a narrow benchmark) or edits any line: asset class, amount, maturity and yield.
2. **Day 0: shocks.** A parallel rate shock reprices bills as zero-coupon bonds and notes as coupon bonds ($P = \Sigma\, C/(1+y)^t + F/(1+y)^n$); gold, bitcoin and equities fall by chosen percentages; secured loans take a credit loss; a bank failure freezes part of the bank cash at an assumed recovery rate.
3. **Days 1-N: the run.** Redemption requests arrive on a front-loaded schedule. The issuer pays $1 per token by selling through a **liquidity waterfall** (cash → repo → money market fund → bills → notes → gold → crypto → other → loans). Each asset can only sell a set share of itself per day and pays a fire-sale discount; requests it cannot meet are queued. Optional **panic feedback** (Diamond & Dybvig, 1983) speeds up the run once reserves fall below par.
4. **Output.** A verdict (HOLDS LIQUIDITY STRAIN BELOW PAR BREAKS), the daily coverage ratio (reserves ÷ tokens), value per token, unmet redemptions, a loss breakdown and a plain-English narrative.

Around this core mechanic: a **Breaking Point** map (441 runs over rate shocks 0-500bp × run sizes 0-100%, giving the exact run size at which the peg fails), a **Reserve Quality** page (duration, a 1-20 day liquidity ladder, a simplified GENIUS Act / MAS eligibility check), a **replay of USDC's March 2023 depeg** and a **live supply / peg monitor** (DefiLlama). A command-line script reproduces any run, and 19 unit tests check the pricing and engine.

Example output (same five standard scenarios for every portfolio)	Verdict	Worst value / token
SVB-style reserves (60% in 10-year notes), +300bp shock, 20% run	BELOW PAR	$0.808
USDT-style reserves, bitcoin −50% and gold −10%, 30% run	BELOW PAR	$0.963
USDC-style reserves, 30% of bank cash frozen with no recovery	BELOW PAR	$0.942
USDC-style reserves, +300bp shock, 20% run	HOLDS	$1.000
USDC on 10 Mar 2023, $3.3bn at SVB lost (market low was about $0.87-0.88)	BELOW PAR	$0.918

2 Who is it for?

Mei, Head of Treasury at a Singapore Major Payment Institution that pays out B2B cross-border invoices to Indonesia, the Philippines and Vietnam. To settle 24/7, the firm pre-funds **US$20-50m of USDC and USDT float**.

Her pain point. Her risk committee wants (a) an exposure limit per stablecoin and (b) a contingency plan for a depeg. Today she has only monthly PDF attestations (a snapshot of what the reserves *were*, published weeks later) and qualitative ratings. Neither tells her **under what conditions this coin stops paying $1 within the five business days MAS expects, or how much float she can safely hold**. The risk is real: in March 2023 a firm holding USDC float watched it fall about 12% over a weekend while US bank rails were closed and redemption was impossible.

Secondary users: crypto exchange and custodian risk teams setting collateral haircuts on stablecoins; analysts and students learning how reserve design drives run risk.

3 Why does it need to exist?

The gap

- **Stablecoins are now payment infrastructure, but their risk is assessed with static disclosures.** USDT and USDC alone back about $257bn of tokens (Tether liabilities $183.6bn; Circle stablecoin holder deposits $72.9bn, 30 Jun 2026). StraitsX alone serves 150+ enterprises (Week 10). Yet holders get monthly or quarterly attestations of composition, not how the reserves *behave* under stress.
- **The failure modes are concrete and different for each coin.** USDC's risk was bank concentration ($3.3bn of about $40bn at SVB). An SVB-style long-duration portfolio fails on rates. A USDT-style mix fails on crypto/gold prices and on liquidity: in our model its secured loans and gold mean only about 48% of tokens can be paid on day 1 and runs above 75% are paid late. One "backing ratio" number hides all of this.
- **Regulation defines safe reserves but does not stress them.** The GENIUS Act (July 2025) and the MAS single-currency stablecoin framework (Aug 2023) list eligible assets (cash, short T-bills of 93 days or less, short repo, government MMFs) and require 100% backing and fast par redemption. Firms still hold coins that sit partly outside these rules (our check: GENIUS-eligible assets cover only about 77% of USDT-style tokens) and need their own view.
- **Existing tools look backwards.** Dashboards track supply and price after the fact; rating agencies give qualitative scores. Neither lets a treasurer ask "what if rates rise 300bp and 30% redeem in five days?"

Why this approach

- **Deterministic, transparent scenario engine** rather than a black-box prediction model: a risk committee must be able to audit every assumption. All liquidity and fire-sale assumptions are visible and editable, and the Breaking Point map shows how sensitive the conclusion is.
- **Borrows proven bank practice** (liquidity stress tests and liquidity ladders) and applies it to stablecoins, using course tools: bond pricing and duration (Week 3), systematic vs idiosyncratic risk (Week 5), runs and stablecoins (Weeks 9-10).
- **Public data only, works offline.** Issuer reports plus free APIs. No keys, no custody, no regulatory licence needed to run it.

4 How could it generate revenue?

Tier	Customer	What they pay for	Indicative price
Free	Anyone	Public scorecard of major stablecoins under standard scenarios: brand and lead generation	$0
Treasury Pro	Payment institutions, fintechs, crypto-native corporates	Custom portfolios and scenarios, exposure-limit reports for risk committees, PDF/CSV exports, alerts when a new attestation shifts a coin's breaking point	US$500-1,500 / month
Enterprise / data licence	Exchanges, custodians, lenders, fund administrators	API of stress metrics per coin (breaking point, day-1 liquidity, eligible backing) to set collateral haircuts and listing limits	US$25-100k / year

Why someone would pay. The value is avoided loss. A 5% depeg on a US$30m float costs US$1.5m. A subscription costing about 1% of that, which also gives the risk committee a documented methodology, is an easy decision. Customers are easy to reach: licensed payment institutions in Singapore are a short list of known firms, and MAS and GENIUS reporting requirements give them a compliance reason to buy. **Risk we would manage:** issuers may offer to pay for "certified" results. We would keep the methodology public and never take issuer fees for scores.

> **Scope and trade-offs.** Built in about four weeks around one reliable core mechanic. Deliberately left out: a secondary-market token price model, user accounts and alerts, yield-curve twists and stochastic simulation, and on-chain reserve proofs. Reserve presets are point-in-time and some maturities/yields are labelled assumptions. This is an educational prototype, not investment or legal advice.

Sources. Circle Internet Group, Form 10-Q, quarter ended 30 Jun 2026 (SEC EDGAR) · Tether Q2 2026 reserves report (BDO Italia), summarised by stablecoininsider.org · CNBC (11 Mar 2023), "Stablecoin USDC breaks dollar peg after firm reveals it has $3.3 billion in SVB exposure" · Paxos, "The GENIUS Act" (2026) · Reed Smith (Aug 2023), "Singapore stablecoin regulation: MAS confirms its framework" · Diamond & Dybvig (1983), Journal of Political Economy 91(3) · DefiLlama stablecoin and price APIs.